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Software Requirements Specification (SRS)

“System Analysis and Design”

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I. Project Charter

Project Title:

SecureMail – Privacy-Focused Secure Email Mobile Application

Project Objective:

To design and develop a mobile-based secure email platform that provides encrypted communication, spam protection, threat detection, and secure account management for users.

SMART Success Criteria:

Specific:

Provide secure login, encrypted messaging, inbox management, and security monitoring.

Measurable:

99% login authentication accuracy

95% spam/threat detection rate

End-to-end encryption for 100% sent messages

Achievable:

Built using existing encryption APIs, SMTP/IMAP protocols, and mobile frameworks.

Relevant:

Addresses privacy, cybersecurity, and secure communication needs.

Time-Bound:

Completion within 4–6 months.

High-Level Risks:

Cybersecurity breaches

Encryption failures

Server downtime

Regulatory compliance issues

User adoption challenges



II. Stakeholders Analysis

Stakeholder	Role	Interest	Influence
End User	Use secure email services	High	High
IT Security Team	Monitor Threats & incidents	High	High
System Admin	Manage System Operations	High	Medium
Developers	Build and Maintain App	High	High
Regulatory Bodies	Data Privacy Compliance	Medium	High

Interest/Influence Matrix:

High Interest / High Influence:

- Users
- IT Security Team
- Developers

High Influence / Low Interest:

- Regulatory Bodies

High Interest / Medium Influence:

- Admin



III. Requirements Elicitation

Methods Used:

User interviews
Security expert consultation
Survey on secure communication needs
Competitive analysis of secure email apps

Key Findings:

Users prioritize privacy and ease of use
2FA is essential
Spam detection is highly valued
Secure file attachments are required
Account recovery is necessary.

IV. Specific Requirements

FUNCTIONAL REQUIREMENTS (FR) — MoSCoW Prioritization

Must Have:

FR1: The system shall allow user registration and login.
FR2: The system shall support password reset and recovery email.
FR3: The system shall implement Two-Factor Authentication (2FA).
FR4: The system shall encrypt all outgoing emails.
FR5: The system shall allow users to send, receive, and organize emails.
FR6: The system shall detect spam and malicious threats.
FR7: The system shall maintain security logs and alerts.



Should Have:

FR8: The system should provide attachment encryption.

FR9: The system should include dashboard analytics.

Should Have:

FR8: The system should provide attachment encryption.

FR9: The system should include dashboard analytics.

Could Have:

FR10: The system could support biometric login.

Won't Have (Current Version):

FR11: Voice-command email composition.

NON-FUNCTIONAL REQUIREMENTS (NFR)

Performance:

NFR1: Login response under 3 seconds

NFR2: Email encryption under 2 seconds

Security:

NFR3: AES/RSA encryption

NFR4: GDPR/Data Privacy compliance

Usability:

NFR5: Mobile-friendly intuitive UI

Reliability:

NFR6: 99.9% uptime



V. Logical Use Case Models

Primary Actors:

- User
- IT Security Team
- Admin

Major Use Cases:

User:

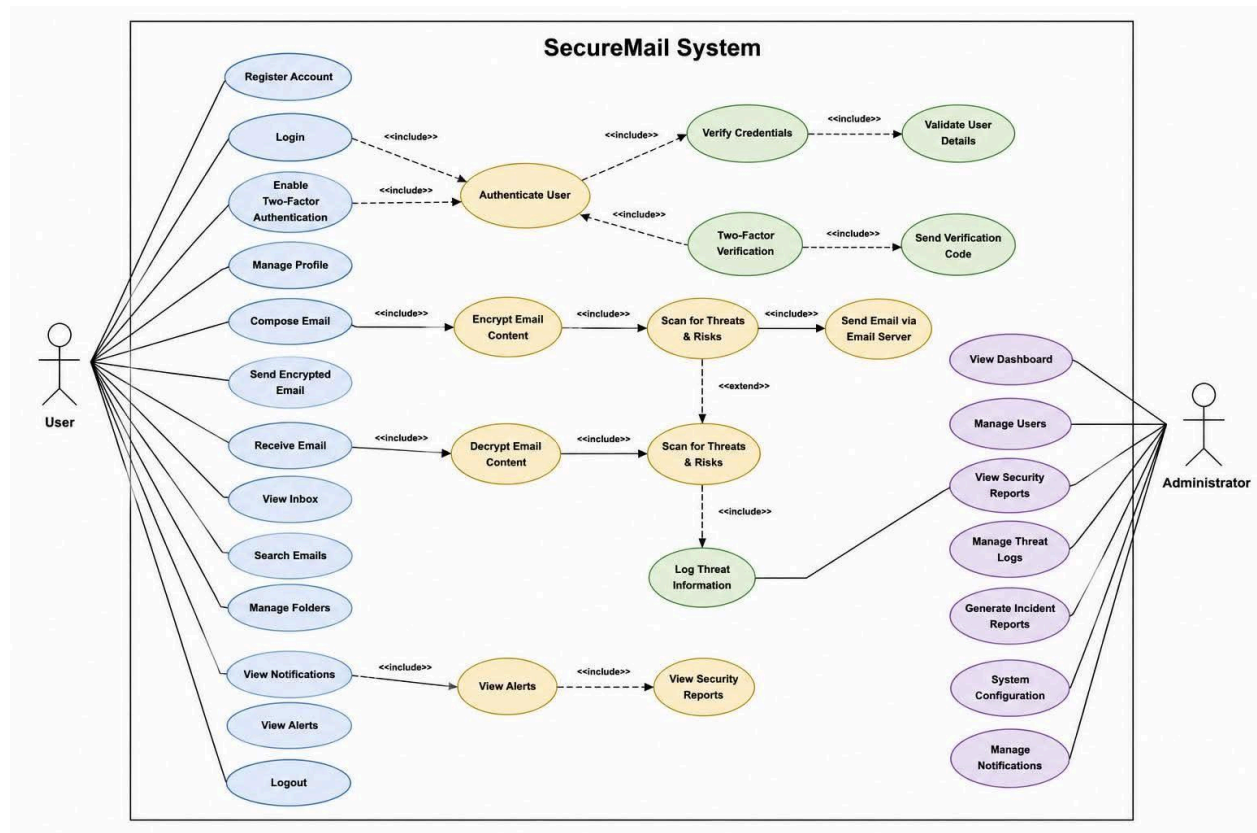
- Register Account
- Login
- Verify 2FA
- Compose Email
- Encrypt & Send Email
- Read Inbox
- Manage Folders
- Reset Password

IT Security Team:

- Monitor Security Alerts
- Review Threat Reports

Admin:

- Manage Accounts
- Review Logs





VI. System Boundary

External Entities:

- User
- IT Security Team
- ADmin
- Authentication Service
- Third-Party Email Server

System Boundary:

SecureMail System

Inputs:

- Login Credentials
- Email Contents
- Monitoring requests

Outputs:

- Secure email
- Alerts
- Reports

